Juyi Lin

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Ph.D. Student, Computer Engineering · Vision-Language-Action Models · Generative World Models · Efficient Multimodal Inference

github.com/LukeLIN-web | Homepage lukelin-web.github.io

Google Scholar scholar.google.com/citations?user=-WMoctgAAAAJ | linkedin.com/in/juyi-lin

Education

Northeastern University · Ph.D. in Computer Engineering · GPA 3.92/4.0 · Advisor: Prof. Yanzhi Wang — Expected 2028.08

King Abdullah University of Science and Technology (KAUST) · M.S. in Computer Science · GPA 3.71/4.0 · Advisor: Prof. Marco Canini — 2022.09 – 2024.05

Zhejiang University · B.Eng. in Electronic Engineering · GPA 3.75/4.0 — 2018.09 – 2022.06

Research & Industry Experience

Futurewei Technologies, Inc. · Research Intern, Generative AI & World Modeling — 2026.05 – 2026.09

LEAP | Learned Block-wise Evidence Retrieval for Long Audio-Video Perception

- **My role:** built LEAP end to end on **Qwen3-Omni-30B-A3B** for hour-scale audio-visual QA — the localization and answer stages, their two LoRAs, and the evaluation harness behind all reported numbers.
- **Method:** per-block localization picks evidence windows; **one bounded answer pass** re-encodes only those windows.
- **Results: +4.5–16.8 pp** over the Qwen3-Omni baseline on TraceAV, LVOmni, VideoOdyssey (1–4 h) and MMOU; transfers to **MiniCPM-o 4.5** (**+3.1–13.0 pp** over its published results); answer input and peak context stay **independent of recording duration**; supports causal/streaming queries without streaming-specific training.

  Paper arxiv.org/abs/2609.39938 · Site lukelin-web.github.io/leap-website

Beijing VirtAI Tech · Infrastructure Engineering Intern — 2022.03 – 2022.07

Distributed Training Performance Measurement System | Framework Integration & Profiling

- **My work:** integrated the **PaddlePaddle** distributed training library and **Deep Graph Library (DGL)** into the in-house performance measurement system, extending it to two new framework backends.
- **Analysis:** profiled PaddlePaddle distributed training and reported its scaling bottlenecks to the platform team.

Selected Projects

Northeastern University · Physical Reasoning Benchmark for Generative World Models — 2025.09 – 2026.03

PhyGround | Criteria-Grounded Benchmark + PhyJudge-9B Open VLM Judge

- **Proposed & built:** a criteria-grounded benchmark for physical reasoning in generative video world models built on **250 curated prompts** and a taxonomy of **13 physical laws** (mechanics, fluids, optics); evaluated **8 video models** against it.
- **Human study:** led a large-scale annotation study — **459 annotators**, **5,796 complete annotations**, **37.4K+ fine-grained labels**, with split-half ranking correlation Spearman's $\rho > 0.90$.
- **Model release:** trained and released **PhyJudge-9B**, an open physics-specialized VLM judge reaching **3.3% aggregate relative bias vs. 16.6% for Gemini-3.1-Pro**.

  Paper arxiv.org/abs/2605.10806 · Code github.com/NU-World-Model-Embodied-AI/PhyGround · Site phyground.github.io · HF huggingface.co/NU-World-Model-Embodied-AI/phyjudge-9B

Northeastern University · Video Reasoning Benchmark 2025.09 – 2026.01

VBVR-EvalKit | A Very Big Video Reasoning Suite for Video Generation Models

- **Proposed & co-led (team of 22):** a reasoning benchmark for video generation models, evaluating **9 models** (4 proprietary, 5 open-weight) — designed the data-generation and benchmark framework, set the roadmap, and contributed core code and evaluation metrics; shipped an open-source ecosystem with **100+ synthetic data generators**.
- **Dataset: 200 curated reasoning tasks** and **1M+ video clips**, 1000× larger than prior datasets; released eval suite = 100 tasks × 5 samples (500 videos).

Paper arxiv.org/abs/2602.20159v2 · Code github.com/Video-Reason/VMEvalKit · Site video-reason.com · HF huggingface.co/VideoReason

Northeastern University, funded by EmbodyX · Vision-Language-Action Models 2025.01 – 2025.09

VOTE | Vision-Language-Action Optimization with Trajectory Ensemble Voting

- **Proposed VOTE:** a fine-tuning framework for parallel VLA action prediction, using an ensemble voting strategy over sampled actions to cut compute while improving success and generalization; **adopted by Cisco and EmbodyX**.
- **Results: +20%** average success over OpenVLA across four LIBERO suites, **+7%** over the state-of-the-art VLA model on SimplerEnv WidowX, and **39×** faster action generation on NVIDIA Jetson Orin.
- **Extension:** pretrained Llama3.2-1B into a VLM and fine-tuned it with VOTE, beating 7B OpenVLA by **16%** average success across the four LIBERO suites.

Paper arxiv.org/abs/2507.05116 · Code github.com/LukeLIN-web/VOTE · Funding embodyx.io

Selected Publications

∗ denotes equal contribution.

LEAP: Learned Block-wise Evidence Retrieval for Long Audio-Video Perception *Preprint*
Juyi Lin, Zhiqiang Lao, Jiali Cui, Lin Zhao, Pu Zhao, Dichang Zhang, Arman Akbari, Yu Qi, and others

PhyGround: Benchmarking Physical Reasoning in Generative World Models *Preprint*
Juyi Lin, Arash Akbari, Yumei He, Lin Zhao, Haichao Zhang, Arman Akbari, Xingchen Xu, Zoe Y. Lu, and others

VOTE: Vision-Language-Action Optimization with Trajectory Ensemble Voting *ICANN 2026*
Juyi Lin, Amir Taherin, Arash Akbari, Arman Akbari, Lei Lu, Guangyu Chen, Taskin Padir, Xiaomeng Yang, and others

A Very Big Video Reasoning Suite *ICML 2026*
Maijunxian Wang∗, Ruisi Wang∗, **Juyi Lin**∗, Ran Ji∗, Thaddäus Wiedemer, Qingying Gao, and others

Audio-Language-Action Models with Direct Speech Conditioning for Robotic Manipulation *DAC 2026 LBR*
Enfu Nan, Pu Zhao, Yixiao Chen, Lin Zhao, **Juyi Lin**, Chen Wang, Weiwei Chen, Yanzhi Wang

Cross-Platform Scaling of Vision-Language-Action Models from Edge to Cloud GPUs *GLSVLSI 2026*
Amir Taherin, **Juyi Lin**, Arash Akbari, Arman Akbari, Pu Zhao, Weiwei Chen, David Kaeli, Yanzhi Wang

ALTER: All-in-One Layer Pruning and Temporal Expert Routing for Efficient Diffusion Generation *NeurIPS 2025*
Xiaomeng Yang∗, Lei Lu∗, Qihui Fan, Changdi Yang, **Juyi Lin**, Yanzhi Wang, Xuan Zhang, Shangqian Gao

Open-Source Acceleration of Stable-Diffusion.cpp Deployable on All Devices *Preprint*
Jingxu Ng, Cheng Lv, Pu Zhao, Wei Niu, **Juyi Lin**, Minzhou Pan, Yun Liang, Yanzhi Wang

Open Source Contributions

ms-swift (ModelScope, 15k+★): a module-level vLLM import in `swift/rollout` crashed every `--use_vllm false` GRPO run on vLLM-free environments; **PR #9498, merged**

Under review, upstream: vLLM-Omni — thinker-only pipeline serving Qwen3-Omni-30B-A3B · **SGLang-Omni** — forward `use_audio_in_video` through the OpenAI HTTP API · **Qwen2.5-Omni** — frame-sharded fix for a `fetch_video` OpenMP-barrier hang

PyTorch Geometric (24k+★): **12 merged PRs** — DDP `NeighborLoader` examples and tutorial, a fix for a distributed-sampling loader that failed to shard data across processes.

Docs & packaging: Diffusers PR #9506 · FlexLLMGen pip release · nndeploy · OSPP 2023

Professional Service

Program Committee: AAAI 2026 / 2027 · WACV 2027 LVR Workshop
Reviewing: ICLR 2027 · NeurIPS 2026 DiffuLM Workshop · ICANN 2026 · EuroSys 2023 Shadow PC
Artifact Evaluation: MLSys 2023 · EuroSys 2024 / 2025 · OSDI 2024 · USENIX ATC 2024

Skills

Languages: Python · C++ · Golang · C · Bash · SQL · MATLAB **Tools:** Git · Docker · LaTeX · Linux
Libraries / Frameworks: PyTorch · Transformers · Diffusers · LoRA · LLaMA-Factory · ms-swift · OpenCompass · NumPy